

Mechanism Black Box Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	mechanism-black-box

Big idea: Scientific explanation often goes beyond prediction by proposing mechanisms.

Overview

Students infer the mechanism inside a sealed system by testing inputs and outputs, then compare competing explanations.

Student Prompt

Inspect Mechanism Black Box Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Mechanism Black Box Stimulus A	Student-facing stimulus 1: Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Mechanism Black Box Stimulus B	Student-facing stimulus 2: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Mechanism Black Box Reveal / Twist	Student-facing stimulus 3: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific explanation often goes beyond prediction by proposing mechanisms.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Mechanism Black Box Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Mechanism Black Box Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	What is added by knowing a mechanism?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how scientific explanation often goes beyond prediction by proposing mechanisms.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use a hidden input-output rule that several mechanisms could initially explain.
- Prepare at least one test input that separates competing explanations.

Ready-to-Use Examples

- Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples.
- Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card.

Suggested Timing

Stage	Teacher move	Watch for
1	Introduce a black box that produces outputs from inputs according to a hidden rule.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups test inputs and record outputs.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students propose a mechanism that could explain the pattern.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups test a new input to discriminate between mechanisms.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why explanation requires more than curve-fitting.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A pattern that predicts is useful, but a mechanism says why.
- What test would make one explanation fail?

Common Misconceptions

- Students may think observation happens before theory in a simple linear order.
- Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

- Names variables, assumptions, controls, anomalies, and limits.
- Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

- Support: use number rules with simple arithmetic.
- Challenge: include noisy data where one output is unreliable.

Extension Tasks

- Students compare mechanism in science with interpretation in history or arts.
- Ask groups to design a black box for another group to investigate.

Related Activities

- Model Failure Challenge
- Anomaly Rescue Mission
- Observation is Theory-Laden